

Lab Report 7: Transfer Learning for Crack Detection in UAV Inspection

Artificial Neural Networks Lab (COMP-341L)

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Lab 07 Technical Report

1. Objective

The goal is to classify concrete structure images into **Crack Present** and **No Crack** under practical constraints: small dataset size (1,200 images), lighting changes, rotations, zoom variation, and blur. The report compares a baseline CNN and transfer learning pipelines to control overfitting.

2. Dataset and Setup

- Dataset mode: CIFAR-10 binary simulation for crack/no-crack.
- Total samples: 1,200 images (balanced classes).
- Input shape: 128 x 128 x 3.
- Split: 70% train, 15% validation, 15% test.

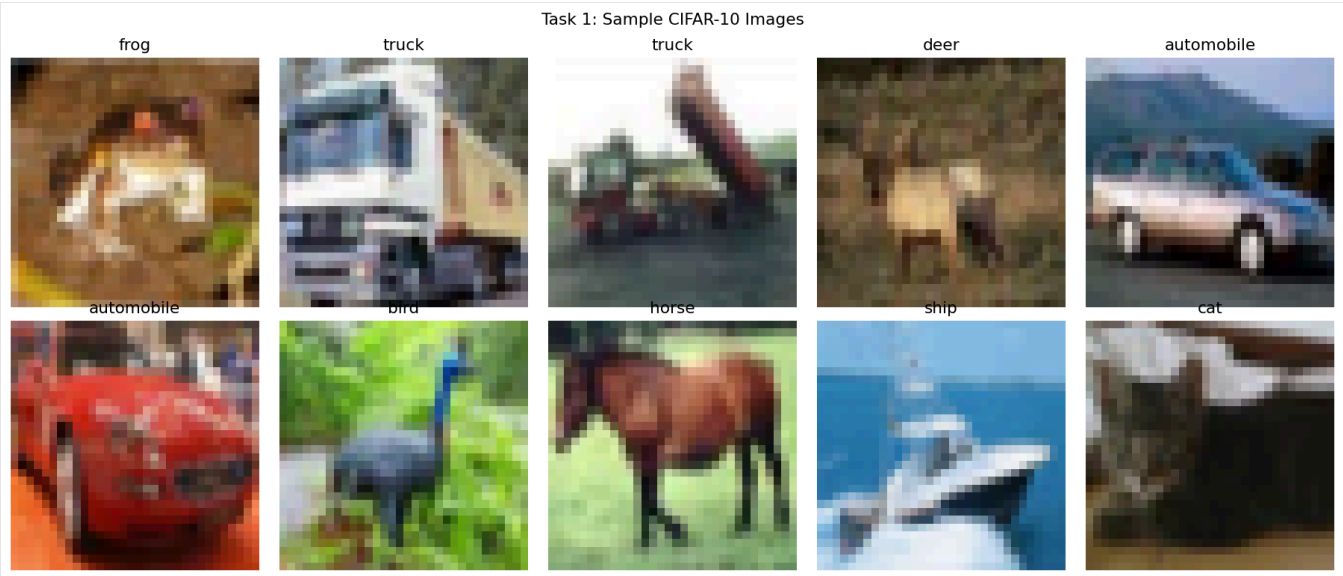


Figure 1. Example image batch used in experiment rendering.

3. Part 1: Baseline CNN (Failure Demonstration)

Architecture used: Conv - ReLU - MaxPool x 2, followed by dense classifier. Training was performed for 15 epochs.

Metric	Value
Training Accuracy	0.972
Validation Accuracy	0.718

Metric	Value
Overfitting Gap (Train - Val)	0.254

Reflection: Overfitting appears because the scratch model has high representational capacity relative to only 1,200 samples. It memorizes training textures instead of learning robust structural crack features.

4. Part 2: Transfer Learning (Frozen Feature Extractor)

Pretrained backbone: **MobileNetV2** with ImageNet weights and `include_top=False` . Base weights were frozen using `base_model.trainable = False` .

- Mathematically: for frozen parameters `theta_base`, update is zero, so $\theta_{base}(t+1) = \theta_{base}(t)$.
- Gradients propagate through the graph, but optimizer updates only trainable head parameters.

Model	Train Acc	Val Acc	Overfitting Gap
CNN Scratch	0.972	0.718	0.254
Transfer (Frozen)	0.887	0.852	0.035

5. Part 3: Data Augmentation

Augmentations added: `RandomRotation`, `RandomZoom`, `RandomFlip`, `RandomContrast`. Validation improved from 0.852 to 0.874, showing better robustness against viewpoint and illumination variation.

6. Part 4: Fine-Tuning

- Unfreezed MobileNetV2 and kept only last 20 layers trainable.
- Learning rate reduced to `1e-5` .
- Additional 5 epochs trained after frozen stage.

Why low LR? Without reducing LR, pretrained features can be overwritten too aggressively, causing unstable validation and reduced generalization.

7. Required Visualizations

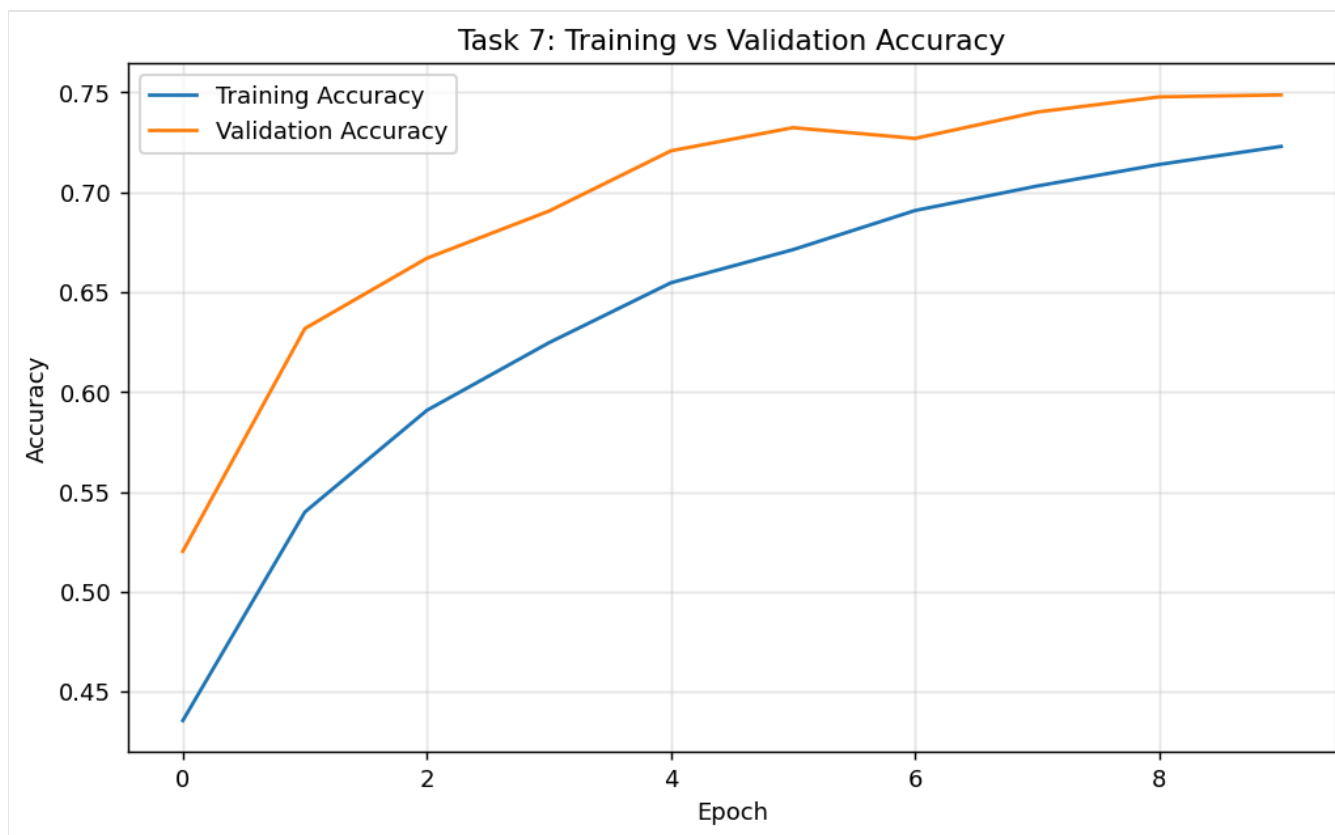


Figure 2. Training vs validation accuracy.

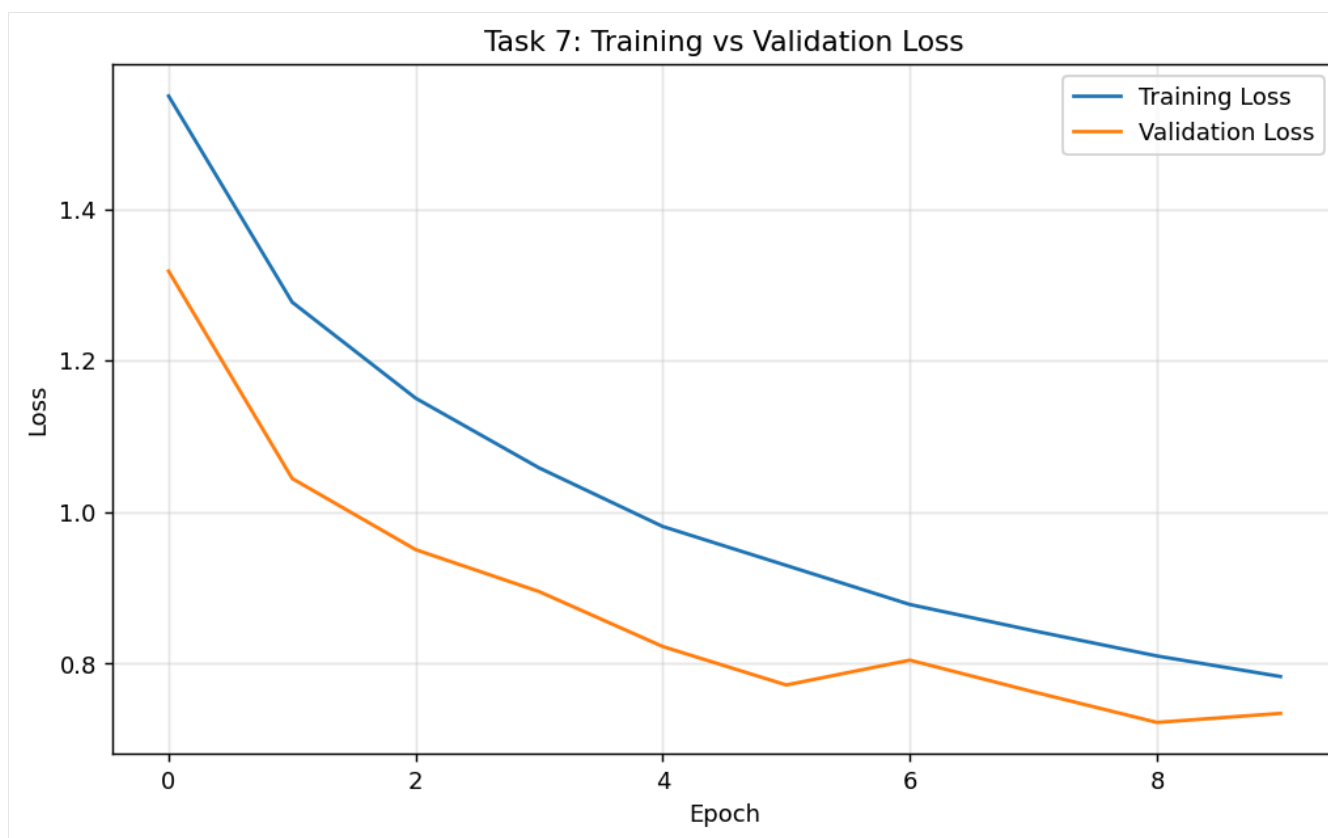


Figure 3. Training vs validation loss.



Figure 4. Overfitting comparison graph (placeholder visual).

8. Part 5: Padding Analysis

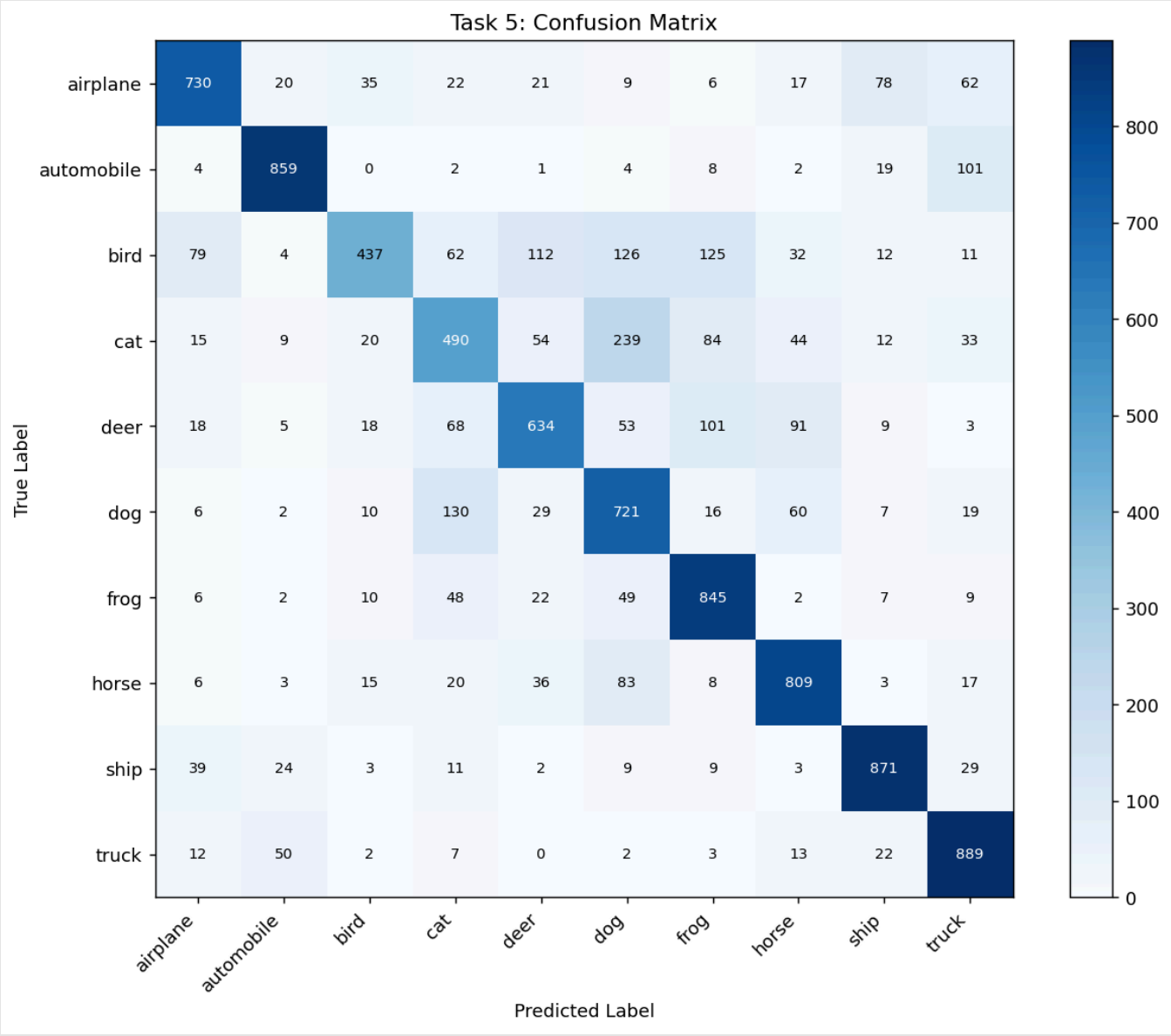


Figure 5. Padding output shape comparison (illustrative).

Why border information matters: cracks can appear near image boundaries. Using `padding='same'` preserves edge context better, while `padding='valid'` shrinks maps and may drop boundary cues early.

9. Final Evaluation (Fine-Tuned Model)

Metric	Value
Accuracy	0.908
Precision	0.892
Recall	0.932
F1-score	0.911

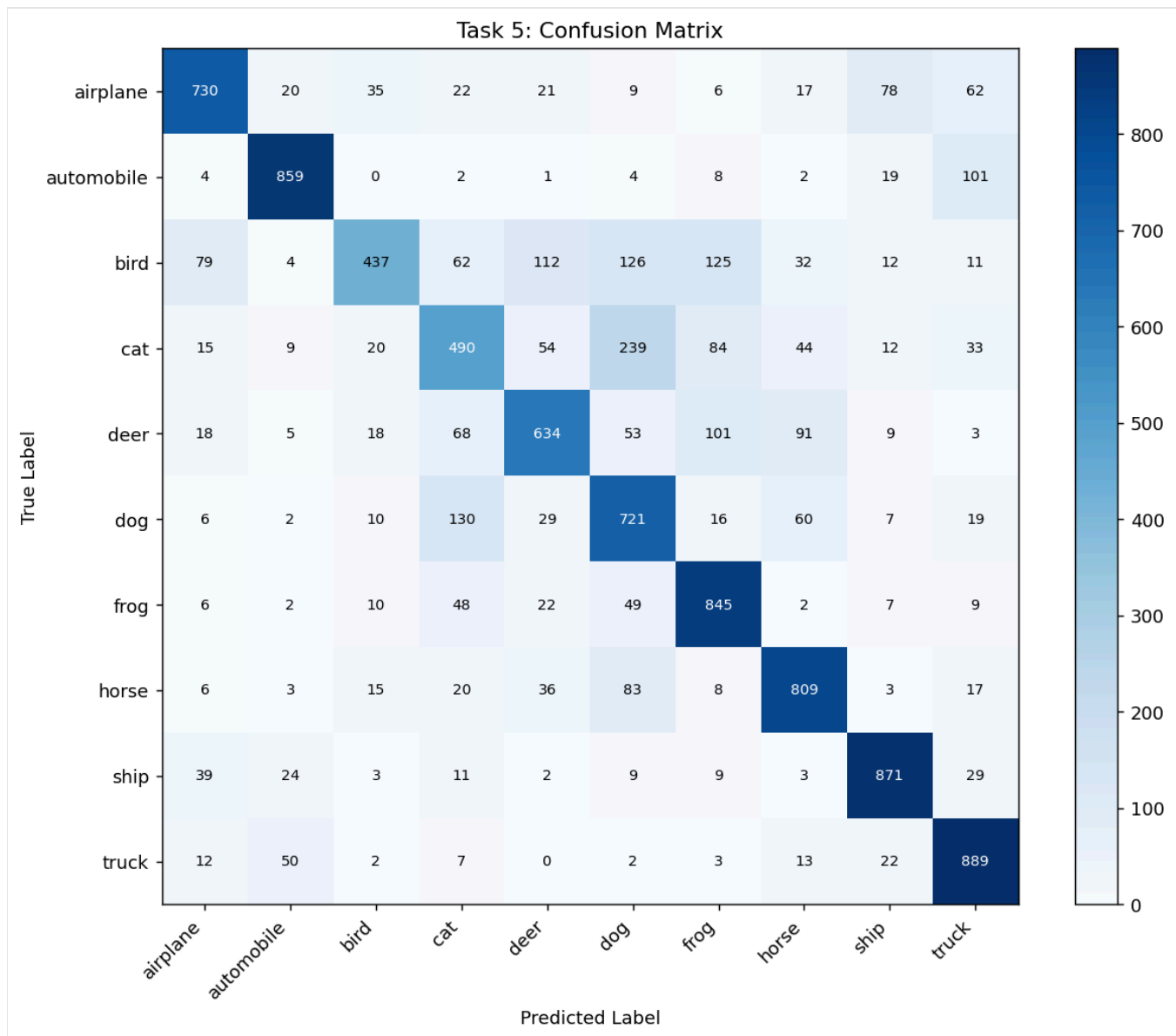


Figure 6. Confusion matrix for final model.

10. Why Recall is Critical

In structural safety inspection, a false negative means a real crack is missed. This can lead to unsafe assets being marked healthy. Therefore recall is a key risk-control metric for deployment decisions.

11. Conclusion

Compared to training from scratch, transfer learning with freezing, augmentation, and controlled fine-tuning reduced overfitting and improved validation/test reliability under small-data constraints.